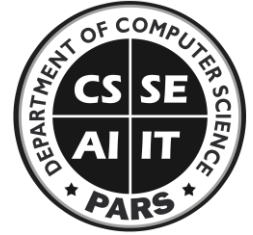




UNIVERSITY OF AGRICULTURE

Department of Computer Science



Lab 3: Programming Fundamentals

(CS-308)

Basic Information			
Lab #:	3	Teacher:	Arsal Mahmood
Objectives	1. Switch Cases in programming		
Registration#	2025-ag-9987	Student Name	Syed Rameez Haider Rizvi
Total Marks	10	Marks Obtained	
Tools	Dev C++ / Visual Studio		

Activity 1 : Previous Lab Evaluation

1. In a company, there are deductions from the salary of the employees for a fund. The deductions rules are as follows:

- If salary is less than Rs.10,000 then no deduction
- If salary is equal to or more than Rs.10,000 and less than Rs.20,000 then deduct Rs.1,000 as fund
- If salary is equal to or more than 20,000 then deduct 7 % of the salary for fund Input salary and name from user and after appropriate deduction show the net payable amount like this:

Hello *name*! Your net payable amount is 12000.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

```
    string name;
```

```
    double salary, netSalary;
```

```
    cout << "Enter your name: ";
```

```

getline(cin, name);

cout << "Enter your salary: ";
cin >> salary;

if (salary < 10000) {
    netSalary = salary;
} else if (salary >= 10000 && salary < 20000) {
    netSalary = salary - 1000;
} else {
    netSalary = salary - (salary * 0.07);
}

cout << "Hello " << name << "! Your net payable amount is " << netSalary << "." << endl;

return 0;
}

```

```

Enter your name: Rameez
Enter your salary: 15000
Hello Rameez! Your net payable amount is 14000.

-----
Process exited after 9.171 seconds with return value 0
Press any key to continue . . .

```

Activity 2 : Switch Statements

2. Develop a calculator which can handle two numbers. The calculator can perform addition, subtraction, multiplication, and division by using switch cases.

```

#include <iostream>
using namespace std;

int main() {
    double num1, num2, result;
    char op;

    cout << "Enter first number: ";
    cin >> num1;

```

```
cout << "Enter second number: ";
```

```
cin >> num2;
```

```
cout << "Enter operation (+, -, *, /): ";
```

```
cin >> op;
```

```
switch(op) {
```

```
    case '+':
```

```
        result = num1 + num2;
```

```
        break;
```

```
    case '-':
```

```
        result = num1 - num2;
```

```
        break;
```

```
    case '*':
```

```
        result = num1 * num2;
```

```
        break;
```

```
    case '/':
```

```
        if(num2 != 0)
```

```
            result = num1 / num2;
```

```
        else {
```

```
            cout << "Error: Division by zero!" << endl;
```

```
            return 0;
```

```
        }
```

```
        break;
```

```
    default:
```

```
        cout << "Invalid operator!" << endl;
```

```
        return 0;
```

```
}
```

```
cout << "Result: " << result << endl;
```

```
return 0;
```

```
}
```

```
Enter first number: 55
Enter second number: 90
Enter operation (+, -, *, /): -
Result: -35

-----
Process exited after 14.42 seconds with return value 0
Press any key to continue . . .
```

3. Write a program that takes grade as an input from the user and depending upon the grade displays the message by using switch. Following are the criteria:

- If grade is 'A' then display “Excellent”
- If grade is 'B' then display “Very Good”
- If grade is 'C' then display “Good”
- If grade is 'D' then display “Poor”
- If grade is 'F' then display “Fail”

```
#include <iostream>
using namespace std;

int main() {
    char grade;

    cout << "Enter your grade (A, B, C, D, F): ";
    cin >> grade;

    switch(grade) {
        case 'A':
            cout << "Excellent" << endl;
            break;
        case 'B':
            cout << "Very Good" << endl;
            break;
        case 'C':
            cout << "Good" << endl;
            break;
        case 'D':
            cout << "Poor" << endl;
            break;
        case 'F':
            cout << "Fail" << endl;
```

```

        break;
    default:
        cout << "Invalid grade!" << endl;
        break;
    }

    return 0;
}

```

```

Enter your grade (A, B, C, D, F): A
Excellent

-----
Process exited after 2.784 seconds with return value 0
Press any key to continue . . .

```

4. Write a program that takes grade as an input from the user and depending upon the grade displays the message by using switch. Now the program will handle uppercase and lowercase letters as well. Following are the criteria:

- If grade is 'A' or 'a' then display “Excellent”
- If grade is 'B' or 'b' then display “Very Good”
- If grade is 'C' or 'c' then display “Good”
- If grade is 'D' or 'd' then display “Poor”
- If grade is 'F' or 'f' then display “Fail”

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    char grade;
```

```
    cout << "Enter your grade (A-F or a-f): ";
```

```
    cin >> grade;
```

```
    switch(grade) {
```

```
        case 'A':
```

```
        case 'a':
```

```
            cout << "Excellent" << endl;
```

```
            break;
```

```
        case 'B':
```

```
        case 'b':
```

```
            cout << "Very Good" << endl;
```

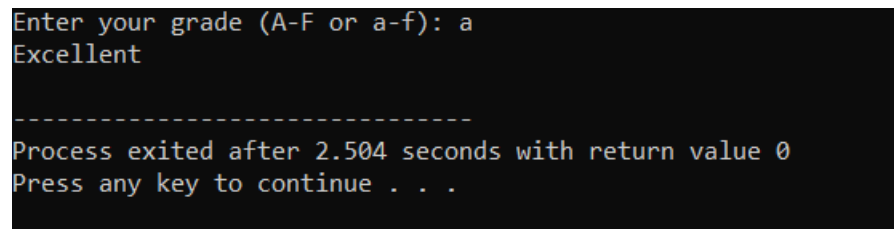
```
            break;
```

```

        case 'C':
        case 'c':
            cout << "Good" << endl;
            break;
        case 'D':
        case 'd':
            cout << "Poor" << endl;
            break;
        case 'F':
        case 'f':
            cout << "Fail" << endl;
            break;
        default:
            cout << "Invalid grade!" << endl;
            break;
    }

    return 0;
}

```



```

Enter your grade (A-F or a-f): a
Excellent

-----
Process exited after 2.504 seconds with return value 0
Press any key to continue . . .

```

5. Input an integer number from user and display weekday name according to the number. As Monday for 1, Tuesday for 2 and so on. The program should display wrong input in case of number less than 1 or greater than 7.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int day;
```

```
    cout << "Enter a number (1-7): ";
```

```
    cin >> day;
```

```
    switch(day) {
```

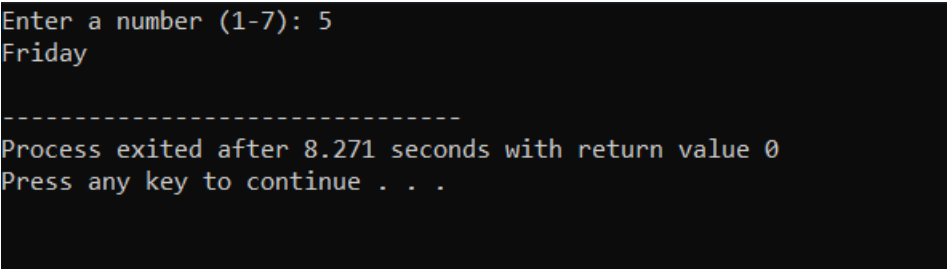
```
        case 1:
```

```

        cout << "Monday" << endl;
        break;
    case 2:
        cout << "Tuesday" << endl;
        break;
    case 3:
        cout << "Wednesday" << endl;
        break;
    case 4:
        cout << "Thursday" << endl;
        break;
    case 5:
        cout << "Friday" << endl;
        break;
    case 6:
        cout << "Saturday" << endl;
        break;
    case 7:
        cout << "Sunday" << endl;
        break;
    default:
        cout << "Wrong input" << endl;
        break;
}

return 0;
}

```



```

Enter a number (1-7): 5
Friday
-----
Process exited after 8.271 seconds with return value 0
Press any key to continue . . .

```

- Write a program that inputs a month number and prints the season it belongs to (Winter, Spring, Summer, Autumn) using switch cases.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {  
    int month;  
  
    cout << "Enter month number (1-12): ";  
    cin >> month;  
  
    switch(month) {  
        case 12:  
        case 1:  
        case 2:  
            cout << "Winter" << endl;  
            break;  
        case 3:  
        case 4:  
        case 5:  
            cout << "Spring" << endl;  
            break;  
        case 6:  
        case 7:  
        case 8:  
            cout << "Summer" << endl;  
            break;  
        case 9:  
        case 10:  
        case 11:  
            cout << "Autumn" << endl;  
            break;  
        default:  
            cout << "Invalid month number!" << endl;  
            break;  
    }  
  
    return 0;  
}
```



```
Enter month number (1-12): 3
Spring

-----
Process exited after 6.649 seconds with return value 0
Press any key to continue . . .
```

7. Input a currency code (USD, EUR, GBP, JPY) and an amount. Use switch-case to print the currency name, country, and conversion to PKR. Include a case for invalid currency codes.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

```
    string code;
```

```
    double amount, converted;
```

```
    cout << "Enter currency code (USD, EUR, GBP, JPY): ";
```

```
    cin >> code;
```

```
    cout << "Enter amount: ";
```

```
    cin >> amount;
```

```
    switch (code[0]) {
```

```
        case 'U': // USD
```

```
        case 'u':
```

```
            if(code == "USD" || code == "usd") {
```

```
                converted = amount * 280; // example rate
```

```
                cout << "Currency: US Dollar" << endl;
```

```
                cout << "Country: USA" << endl;
```

```
                cout << "Amount in PKR: " << converted << endl;
```

```
                break;
```

```
            }
```

```
        case 'E': // EUR
```

```
        case 'e':
```

```
            if(code == "EUR" || code == "eur") {
```

```
                converted = amount * 310; // example rate
```

```
                cout << "Currency: Euro" << endl;
```

```
                cout << "Country: European Union" << endl;
```

```

        cout << "Amount in PKR: " << converted << endl;
        break;
    }
case 'G': // GBP
case 'g':
    if(code == "GBP" || code == "gbp") {
        converted = amount * 360; // example rate
        cout << "Currency: British Pound" << endl;
        cout << "Country: United Kingdom" << endl;
        cout << "Amount in PKR: " << converted << endl;
        break;
    }
case 'J': // JPY
case 'j':
    if(code == "JPY" || code == "jpy") {
        converted = amount * 2.1; // example rate
        cout << "Currency: Japanese Yen" << endl;
        cout << "Country: Japan" << endl;
        cout << "Amount in PKR: " << converted << endl;
        break;
    }
default:
    cout << "Invalid currency code!" << endl;
    break;
}

return 0;
}

```

```

Enter currency code (USD, EUR, GBP, JPY): usd
Enter amount: 10
Currency: US Dollar
Country: USA
Amount in PKR: 2800

-----
Process exited after 5.277 seconds with return value 0
Press any key to continue . . .

```

8. We want to simulate a very basic ATM system that does the following:
Ask the user to enter a PIN.

If the PIN is correct, show a menu:

- 1 → Check Balance
- 2 → Withdraw Money
- 3 → Deposit Money
- 4 → Exit

Based on the menu choice, perform the corresponding action.

Handle wrong PIN and invalid menu choices with error messages.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    const int correctPIN = 1234;
```

```
    int enteredPIN;
```

```
    int choice;
```

```
    double balance = 50000; // initial balance
```

```
    double amount;
```

```
    cout << "Enter your PIN: ";
```

```
    cin >> enteredPIN;
```

```
    if (enteredPIN != correctPIN) {
```

```
        cout << "Wrong PIN! Access denied." << endl;
```

```
        return 0;
```

```
    }
```

```
    cout << "PIN accepted. Welcome!" << endl;
```

```
    cout << "ATM Menu:" << endl;
```

```
    cout << "1 → Check Balance" << endl;
```

```
    cout << "2 → Withdraw Money" << endl;
```

```
    cout << "3 → Deposit Money" << endl;
```

```
    cout << "4 → Exit" << endl;
```

```
    cout << "Enter your choice: ";
```

```
    cin >> choice;
```

```
switch(choice) {
    case 1:
        cout << "Your balance is: " << balance << endl;
        break;
    case 2:
        cout << "Enter amount to withdraw: ";
        cin >> amount;
        if(amount > balance) {
            cout << "Insufficient balance!" << endl;
        } else {
            balance -= amount;
            cout << "Withdrawal successful. New balance: " << balance << endl;
        }
        break;
    case 3:
        cout << "Enter amount to deposit: ";
        cin >> amount;
        balance += amount;
        cout << "Deposit successful. New balance: " << balance << endl;
        break;
    case 4:
        cout << "Exiting. Thank you!" << endl;
        break;
    default:
        cout << "Invalid menu choice!" << endl;
        break;
}

return 0;
}
```

```
Enter your PIN: 1234
PIN accepted. Welcome!
ATM Menu:
1 ? Check Balance
2 ? Withdraw Money
3 ? Deposit Money
4 ? Exit
Enter your choice: 2
Enter amount to withdraw: 15000
Withdrawal successful. New balance: 35000

-----
Process exited after 12.47 seconds with return value 0
Press any key to continue . . .
```

{end of Lab}